

**Test 1- Number and Probability**



**Baldivis**  
Secondary College

No resources allowed

Name: P. Answers

Class: \_\_\_\_\_

Time: 15 minutes

Total Mark: \_\_\_\_\_

1. A letter is selected at random from the word STATISTICS. The probability of selecting a vowel is:

A  $\frac{1}{5}$

B  $\frac{1}{4}$

**C  $\frac{3}{10}$**

D  $\frac{1}{3}$

E  $\frac{2}{5}$



[1 mark]

2. A number is chosen from the set of positive single-digit integers; that is, integers between 1 and 9 inclusive. If A is the set of odd numbers between 1 and 9 inclusive and B is the set of prime numbers between 1 and 9 inclusive:

List the following sets.

i The sample space

1, 2, 3, 4, 5, 6, 7, 8, 9



ii A

1, 3, 5, 7, 9



iii B

2, 3, 5, 7



[3 marks]

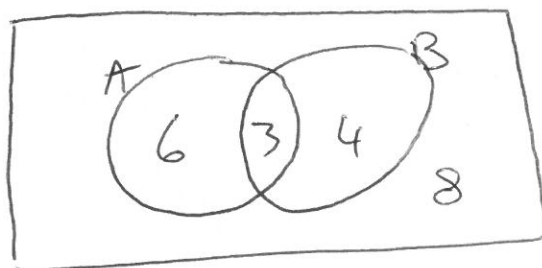
Questions 3 to 6 refer to the following two-way table:

|       | A | Not A | Total |
|-------|---|-------|-------|
| B     | 3 | 4     | 7     |
| Not B | 6 | 8     | 14    |
| Total | 9 | 12    | 21    |



3. Complete the two way table.

4. Represent the information from the two way table in a Venn diagram.



✓  
✓  
✓

5. Pr (Not A) is equal to:

A  $\frac{6}{9}$

B  $\frac{8}{14}$

C  $\frac{7}{21}$

D  $\frac{9}{21}$

E  $\frac{12}{21}$

✓

6. Pr(not A or B) is equal to:

A  $\frac{6}{21}$

B  $\frac{8}{21}$

C  $\frac{7}{21}$

D  $\frac{11}{21}$

E  $\frac{3}{21}$

✓

[1 + 3 + 1 + 1 = 6 marks]

7.  $(2x)^4$  in expanded form is:

A  $x \times x \times x \times x$

B  $2x^4 \times 2x^4 \times 2x^4 \times 2x^4$

C  $(2x)^4 \times (2x)^4 \times (2x)^4 \times (2x)^4$

D  $2 \times x \times x \times x \times x$

E  $2 \times 2 \times 2 \times 2 \times x \times x \times x \times x$

[1 mark]

8. The number of significant figures given in 0.00703 is: (Circle the correct answer)

A two

B three

C four

D five

E six

[1 mark]

9.  $\frac{Aa^5b}{8a^3b^2}$  simplifies to: (Circle the correct answer)

A  $\frac{a^5}{2}$

B  $\frac{2a^5}{b}$

C  $\frac{a^5}{2b}$



D  $\frac{2a^5}{b^3}$

E  $\frac{a^5}{2b^3}$

[1 mark]

10. Evaluate each of the following.

a  $6^{-2}$   $\frac{1}{36}$

b  $5^{-3}$   $\frac{1}{125}$



[2 marks]

[Total 15 marks]

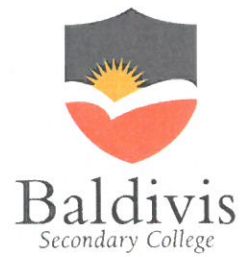
End of Paper.



# Year 9

## Test 1- Number and Probability

Resources allowed



Name: \_\_\_\_\_ Class: \_\_\_\_\_ Time: 35 minutes Total Mark: \_\_\_\_\_

1. Two people are selected without replacement from a group of four: Adam (A), Brenda (B), Caroline (C) and Darren (D).

- a List all the possible combinations for the selection by completing the following table.

|     |   | 1st |    |    |    |
|-----|---|-----|----|----|----|
|     |   | A   | B  | C  | D  |
| 2nd | A | AA  | AB | AC | AD |
|     | B | BA  | BB | BC | BD |
|     | C | CA  | CB | CC | CD |
|     | D | DA  | DB | DC | DD |

- b Find the probability that the selection will contain Brenda and Darren.

$$\frac{2}{16} = \frac{1}{8}$$

- c Find the probability that the selection will contain Caroline.

$$\frac{7}{16}$$

- d Find the probability that the selection will be the same person both times.

$$\frac{4}{16} = \frac{1}{4}$$

- e Find the probability that Adam won't be selected.

$$\frac{9}{16}$$

[5 marks]

2. Simplify each of the following ratios with correct units.

a 180 : 30

$$6 : 1$$

b 450 m : 90 cm

$$\begin{aligned} 450 : 0.9 \\ 4500 : 9 \\ 500 : 1 \end{aligned}$$

c 63 : 7

$$9 : 1$$

d 8 km : 2000 cm : 300 m

$$8 : 0.02 : 300$$

$$\begin{aligned} 8000 : 20 : 300 \\ 800 : 2 : 30 \\ 400 : 1 : 15 \end{aligned}$$

[2 marks]

3. 300 Postcards were chosen at random. 205 postcards had correctly written postcodes on them.

- a Use the above information to calculate the experimental probability of choosing a postcard that has a correctly written postcode on it.

$$\frac{205}{300}$$

- b Calculate the experimental probability of choosing a postcard that has an incorrectly written postcode on it

$$\frac{95}{300}$$

- c Find the expected number of postcards with correctly written postcodes from the following group sizes:

- i 800 postcards

$$\frac{205}{300} \times 800$$

$$= 546$$

- ii 5500 postcards

$$3758$$

[1 + 1 + 2  $\times$   $\frac{1}{2}$  = 3 marks]

4. Round the following to 3 decimal places.

- a. 67.3957

$$67.396$$

- b. 299.8977

$$299.898$$

[1 mark]

5. Ben rides his bike for  $3\frac{1}{2}$  minutes and travels 700 m. What is his average speed? (Circle the correct answer)

A 150 m/min

**B 200 m/min**

C 250 m/min

D 300 m/min

E 350 m/min

[1 mark]

6. As a decimal,  $5\frac{7}{12}$  is equal to: (Circle the correct answer)

A 5.583

B 5.583

C  $\overline{5.583}$

**D  $\overline{5.583}$**

E 5.583

[1 mark]

7. Which of the following has the largest value? (Circle the correct answer)

A 0.04

B  $\frac{4}{11} 0.\overline{36}$

C 30% 0.3

D  $\frac{3}{8} 0.375$

E 0.3

[1 mark]

8. Evaluate the following, giving your answer as a mixed number when appropriate.

a  $\frac{7}{10} + \frac{1}{10}$

$\frac{8}{10} = \frac{4}{5}$

b  $\frac{2}{3} \div \frac{8}{9}$

$\frac{2}{3} \times \frac{9}{8} = \frac{18}{24} = \frac{3}{4}$

c  $2\frac{1}{2} - 1\frac{5}{6}$

$\frac{5}{2} - \frac{11}{6} \Rightarrow \frac{15}{6} - \frac{11}{6} = \frac{4}{6} = \frac{2}{3}$

d  $1\frac{7}{8} \times 3\frac{1}{5}$

$\frac{15}{8} \times \frac{16}{5} = \frac{240}{40} = 6$

[4 marks]

9.  $3x^5 \times 6x^3$  is equivalent to: (Circle the correct answer)

A  $18x^8$

B  $18x^{15}$

C  $18x^2$

D  $9x^{15}$

E  $9x^8$

[1 mark]

10. Express each of the following using positive indices only.

a  $x^{-1}$

$\frac{1}{x}$

b  $\frac{1}{y^{-6}}$

$y^6$

c  $6a^{-3}b$

$\frac{6b}{a^3}$

d  $\frac{4x^4}{2^{-1}y^{-2}}$

$8x^4y^2$

$[\frac{1}{2} + \frac{1}{2} + 1 + 1 = 3 \text{ marks}]$

11. The figure 2613.23 rounded to 3 significant figures is: (Circle the correct answer)

A 3000

**B 2610**

C 2613.230

D 2600.00

E 2613

[1 mark]

12. Simplify each of the following.

a  $(w^2)^4$

$w^8$

b  $\frac{2x^4y^3 \times 3x^5y}{x^2y^2} = \frac{6x^9y^4}{x^2y^2}$

$= 6x^7y^2$

c  $t^4 \times t^5$

$t^9$

d  $\frac{y^7}{y^3}$

$y^4$

[2 marks]

13. Evaluate the following.

a  $(-6y)^0$

1

b  $-4z^0$

-4

[2 marks]

14. Andrew owns a greengrocer's store. Every morning he goes to the market and purchases fruit and vegetables, which he then sells at his store.

When at the market one morning, Andrew sees a 6 kg crate of oranges for \$7.50 and an 8 kg crate of oranges for \$9.20. Which is better value?

~~$6 \div 7.5 = 0.8$~~

~~$8 \div 9.2 = 0.87$~~

$7.5 \div 6 = 1.25 \text{ \$/kg}$

$9.2 \div 8 = 1.15 \text{ \$/kg}$

8kg at \$9.20 is better value

[2 marks]

15.

a. The diameter of a nitrogen atom is measured as 0.000000065 mm in one experiment. Write this measurement using scientific notation.

$6.5 \times 10^{-8}$

b. The diameter of a sodium atom is measured as  $1.8 \times 10^{-7}$  mm. Write this measurement in decimal form.

~~0.00000018~~

0.00000018

- c. The volume of the Moon is 21960000000  $\text{km}^3$ . Write this measurement using scientific notation.

$$2.196 \times 10^{10} \text{ km}^3$$

- d.  $1.2 \times 10^7 \times 3.5 \times 10^6 =$

$$4.2 \times 10^{13}$$

[2 marks]

16. A spinner numbered 1–100 (assume chance of each number is equal) is spun twice. Find

- a Pr(sum of 2 spun = 200)

$$\frac{1}{10000}$$

- b Pr(sum of 2 spun = 3)

$$\frac{2}{10000} = \frac{1}{5000}$$

- c Pr(same number spun twice)

$$\frac{100}{10000} = \frac{1}{100}$$

[1 + 1 + 2 = 4 marks]

[Total 35 marks]

